

L4.3 Latency and Availability Attacks for AD

Quiz

1. What is the central difference between integrity attacks (L4.1, L4.2) and availability / latency attacks (L4.3)?

- A) Latency attacks often leave the output unchanged but late, so accuracy monitors stay silent
- B) Latency attacks always change the predicted class, so accuracy monitors fire at once
- C) Latency attacks work only on cameras, while integrity attacks work on every sensor
- D) Latency attacks require that the attacker knows the ground-truth labels of the scene

ANS:

2. A perception pipeline is delayed by 200 ms while the vehicle drives at 30 m/s. How far does it travel in that extra time?

- A) 0.6 m
- B) 6 m
- C) 60 m
- D) 15 m

ANS:

3. What is 'intermediate-work amplification'?

- A) An input that is much larger in size than usual, such as a huge high-resolution image
- B) A normal-sized input drives many more boxes, tracks or tokens through a costly stage
- C) A flood of requests that exhausts the bandwidth and the request queue of a server
- D) An increase of the learning rate of the victim model during its online training

ANS:

4. The survey defines the amplification factor as $AF = M(x_{adv}) / \text{median } M(x_0)$. What must be held constant between the attacked and the benign measurements?

- A) The attacker's perturbation budget and the number of attack iterations
- B) The random seed that was used for generating the adversarial examples
- C) The model, batch size, hardware and concurrency
- D) The operating system version that was installed on the attacker's machine

ANS:

5. Why does the survey prefer the absolute deadline-miss increase ($p_{attack} - p_{clean}$) to a ratio of miss rates?

- A) Ratios cannot be computed correctly in floating-point arithmetic on a modern GPU
- B) Absolute numbers are easier to plot in a figure than ratios of two numbers are
- C) Clean systems have a near-zero miss rate, so a ratio is undefined or meaningless
- D) Ratios cannot be compared across different GPUs and different software versions

ANS:

6. A job has absolute deadline $d = 45$ ms and completes at $f = 60$ ms. What are its slack $S = d - f$ and tardiness $\tau = \max(0, f - d)$?

- A) $S = 15$ ms, $\tau = 0$ ms
- B) $S = 105$ ms, $\tau = 60$ ms
- C) $S = -15$ ms, $\tau = 0$ ms
- D) $S = -15$ ms, $\tau = 15$ ms

ANS:

7. An EOS-delaying prompt, a no-EOS backdoor and an EOS-suppressing bit flip all lengthen the generated text. In the survey's terms, how do they relate?

- A) Three different bottlenecks that are reached through one common delivery channel
- B) Same bottleneck (output length), three delivery channels; one hard token cap bounds all

- C) Three attacks that cannot be bounded by one common control mechanism
- D) Not latency attacks at all, because every one of them changes the final answer

ANS:

8. Why do many small, spread-out phantom boxes make greedy NMS slow?

- A) They overlap nothing, so almost none is suppressed and the work nears n^2 comparisons
- B) They raise the image resolution, so that the detector head needs much more compute
- C) They force the detector to run a second time on the same input image again
- D) They are suppressed at once, but sorting all of them is the expensive part

ANS:

9. In the Groundswell example, a $188\times$ rise in NMS candidates gives only a $2.7\times$ rise in end-to-end latency. Why?

- A) NMS is a small part of the benign latency, so the other stages dilute the effect
- B) The attack was run on a slower GPU than the one that was used for the clean runs
- C) End-to-end latency is always measured in frames and not in milliseconds
- D) The tracker removed most of the candidates before NMS could process them at all

ANS:

10. What does the survey conclude about the evidence level of physical latency attacks (e.g., SlowPerception, DetStorm)?

- A) All of them were demonstrated end to end on public roads with real traffic
- B) None of them was ever evaluated in a simulator or on target hardware
- C) Physical delivery is shown only in pieces; no entry is a full on-road deployment
- D) Only digital attacks were ever measured on real embedded hardware

ANS:

11. What is a key finding of the EVADE study (Monteuuis et al., NeurIPS 2025) on latency NMS attacks?

- A) Every NMS attack transfers unchanged to all detectors and all hardware platforms
- B) GPU NMS can always be pushed beyond a 500 ms budget by a small patch
- C) None of the three tested defenses reduced the effect on post-NMS counts
- D) Attacks do not transfer across YOLO versions or sizes, and universal patches fail across datasets

ANS:

12. What is the key difference between the NMS bottleneck and the tracker bottleneck?

- A) The tracker runs before the detector, so it cannot see any of the phantom boxes
- B) NMS keeps its state across frames, while the tracker is reset in every frame
- C) The cost of the tracker is always linear in the size of the input image
- D) The tracker has memory: unmatched false tracks survive for frames, so work accumulates

ANS:

13. Why does a per-frame cap on detections not stop the tracker-pool accumulation of SlowTrack?

- A) It bounds one dimension of the $M \times N$ cost, not the persistent pool of tracks
- B) Caps of this kind only work in simulation and never on real embedded hardware
- C) The cap is applied only after the tracker has already finished its full update
- D) SlowTrack never creates more than one single false track in each frame

ANS:

14. Exercise: $T = 30$ ms, deadline $D = 45$ ms, the tracker has share $f = 5\%$ and is amplified by $A = 50$ ($A_{\text{eff}} = 1 + f(A - 1)$). What is the new pipeline latency, and is the deadline missed?

- A) About 45 ms; no, the deadline is met exactly
- B) About 1,500 ms; yes, the deadline is missed
- C) About 36 ms; no, the deadline is met
- D) About 103.5 ms; yes, the deadline is missed

ANS:

15. Continuing the exercise ($T = 30$ ms, $D = 45$ ms, $f = 5\%$), an admission cap limits the amplification to $A = 5$. What is the result?

- A) $A_{e2e} = 5$, latency 150 ms; the deadline is missed
- B) $A_{e2e} = 3.45$, latency 103.5 ms; the deadline is missed
- C) $A_{e2e} = 1.2$, latency 36 ms; the deadline is met
- D) $A_{e2e} = 0.2$, latency 6 ms; the pipeline would be idle

ANS:

16. Why does TrackFlood work against detectors without NMS (YOLOv10, RT-DETR)?

- A) These detectors have no detection head, so the tracker receives no input
- B) The detector latency grows by $100\times$, while the tracker stays unchanged
- C) Detectors without NMS cannot be given an adversarial perturbation
- D) The bottleneck moves to the tracker, so NMS-centric defenses have nothing to act on

ANS:

17. What is the difference between an admission cap and an emission cap?

- A) Admission limits what leaves a stage, while emission limits what enters that stage
- B) Both bound the cost of the stage in exactly the same way for every detector
- C) Admission limits work entering a stage; emission (max_det) limits only its output
- D) Emission caps are used only for tokens and admission caps only for boxes

ANS:

18. Which control on reasoning length gives a hard guarantee according to the survey?

- A) Runtime termination or token accounting that enforces a configured upper bound
- B) A prompt instruction that politely asks the model to think only briefly
- C) A learned length direction that shortens the answers on average over many prompts
- D) A detector that flags verbose prompts before they reach the reasoning model

ANS:

19. Why does the lecture say that AI-based perception has no WCET (worst-case execution time)?

- A) Neural networks run in constant time on every possible input that they receive
- B) Run time depends on the input, and attacks find high-cost inputs, not proven maxima
- C) WCET analysis is forbidden for machine-learning components by the ISO 26262 standard
- D) GPUs cannot be timed accurately by any measurement tool that is available today

ANS:

20. Two vehicles suffer the same perception delay. Vehicle 1 keeps driving on stale data; vehicle 2 performs a safe stop. According to condition C6 (consequence mapping), what is the likely difference?

- A) Both vehicles crash, because the delay is identical in the two cases
- B) Vehicle 1 may crash, while vehicle 2 loses availability but avoids a collision
- C) Vehicle 2 crashes, because stopping on the road is always an unsafe action
- D) Neither vehicle is affected, because latency attacks do not change any behaviour

ANS: